

Digital Electronics

CHAPTER 2

Boolean Algebra

Neso Academy

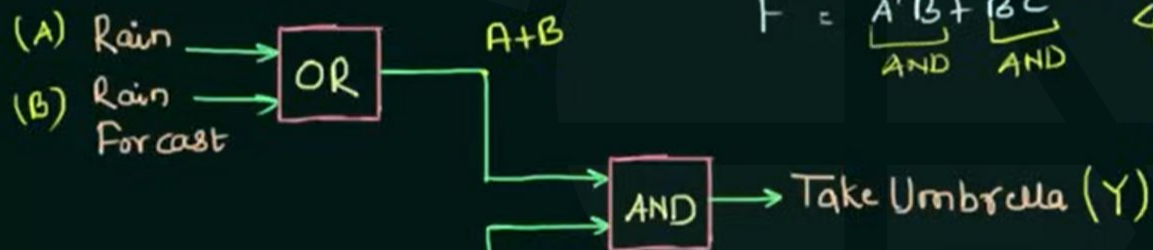


Introduction to Boolean Algebra

>> It is the set of rules used to simplify the given logic expression without changing its functionality.

>> It is used when number of variables are less.

Map method (1,2,3)



(C) Take Can →

NOT

c'

$$Y = (A+B) \cdot c'$$
$$Y = BA'c' + B'A'c' + Bc'$$

5 gates

$$F = \overline{A}B + BC + ABC$$

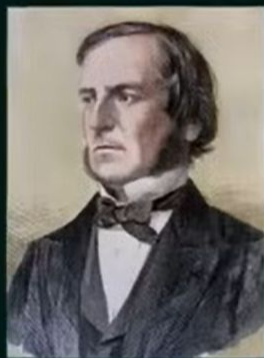
1

$$F = \underbrace{\overline{A}B}_{\text{AND}} + \underbrace{BC}_{\text{AND}} + \underbrace{ABC}_{\text{AND}}$$

11

$$F = \underbrace{\overline{A}B}_{\text{AND}} + \underbrace{BC}_{\text{AND}}$$

4 gates



George Boole
(1815-1864)

Rules: - i) Complement: -

A complement = $\bar{A}$ or A' or (not A)

$$* (A')' = A$$

$0' = 1$
$1' = 0$

$$0' = 1' = 0 \equiv (0')'$$

ii) AND

v.v. imp

$$A \cdot A = A$$

$$A \cdot 0 = 0$$

$$A \cdot 1 = A$$

$$A \cdot A' = 0$$

I/P		O/P ↑
A	B	Y
0	0	0
0	1	0
1	0	0
1	1	1

$$2^2 = 4$$



$$A = 0 \rightarrow 0 \cdot 0 = 0 = Y$$

$$A = 1 \rightarrow 1 \cdot 1 = 1 = Y$$

$$1 \cdot 0 = 0$$

$$0 \cdot 0 = 0$$

$$A = 0 \Rightarrow A' = 1$$

$$A \cdot A' = 0 \cdot 1 = 0$$

$$A = 1 \Rightarrow A' = 0$$

$$A \cdot A' = 1 \cdot 0 = 0$$

iii) OR

$$A + A = A$$

$$A + 0 = A$$

$$A + 1 = 1$$

$$A + A' = 1$$

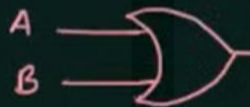
$$A = 0$$

$$0 + 1 = 1$$

$$A = 1$$

$$1 + 1 = 1$$

A	B	Y
0	0	0
0	1	1
1	0	1
1	1	1



$$Y = A + B$$

$$X + 1 = 1$$

$$Y = A'BC + B'CA + A'B + AB + 1$$

$$Y = 1$$

$A = 0$ $A + A'$ $0 + 1 = 1$	$A = 1$ $A + A'$ $1 + 0 = 1$
------------------------------------	------------------------------------

$$A = 0$$

$$Y = 0 + 0 = 0$$

$$A = 1$$

$$Y = 1 + 1 = 1$$

iv) Distributive law: -

V.V.V. imp

$$A + \bar{A}B = A + B$$

$$\bar{A} + AB = \bar{A} + B$$

$$A(B+C) = AB + AC$$

V.V.V. imp

$$A(B+C) = \underline{(A+B) \cdot (A+C)}$$

$$\begin{aligned} A + \bar{A}B &= (A + \bar{A}) \cdot (A + B) \\ &= 1 \cdot (A + B) \\ &= 1 \cdot X \\ &= X \\ &= A + B \end{aligned}$$

v) Commutative Law: -

$$A + B = B + A \quad | \quad A \cdot B = B \cdot A$$

vi) Associative law

$$(A \cdot B) \cdot C = A \cdot (B \cdot C)$$

Priority :-

NOT
AND
OR

vii) De Morgans law :-

$$\overline{(A+B)} = A' \cdot B'$$

$$\overline{(A \cdot B)} = A' + B'$$

$$Y = (A+B) \cdot C' \quad \leftarrow A' + B'$$

$$Y = BA' + B'A' + BC'$$

$$Y = B[A'] + B'[A'] + BC'$$

$$Y = A' [B + B'] + BC'$$

$$Y = \frac{A'}{X} \cdot 1 + BC'$$

$$X \cdot 1 = X \quad \rightarrow \quad Y = X \cdot 1 + BC'$$

$$= X + BC'$$

$$Y = A' + BC'$$

Boolean Algebra Examples (Part 1)

Ex. 1: $AB + AB'$

Ex. 2: $AB + AB'C + AB'C'$

Rules: -

NOT

1. Compliment: $(A')' = A$

AND
OR

2. AND:
 $A.A = A$
 $A.0 = 0$
 $A.1 = A$
 $A.A' = 0$

3. OR:
 $A+A = A$
 $A+0 = A$
 $A+1 = 1$
 $A+A' = 1$

4. Distributive: $A+BC = (A+B).(A+C)$
 $A.(B+C) = A.B + A.C$

* $A+A'B = A+B$
 $A'+AB = A'+B$

5. De Morgan's Law:

$$(A+B)' = A'.B'$$

$$(A.B)' = A'+B'$$

Boolean Algebra Examples (Part 1)

Ex. 1: $AB + AB' = A$

Sol: $F = A \cdot B + A \cdot B'$

$$F = A \cdot [B + B']$$

$$F = A \cdot 1$$

$$\boxed{F = A}$$

Ans: $\begin{matrix} 1 \cdot 1 & 1 \cdot 0 & + & 1 \cdot 1 & + & 1 \cdot 0 \\ 0 & + & 1 & = & 1 \end{matrix}$

$$1 \cdot 0 + 1 \cdot 1,$$

A	B	$F = AB + AB'$
0	0	0
0	1	0
1	0	1
1	1	1

$F = A$

Ex. 2: $AB + AB'C + AB'C' = A$

Sol: $F = AB + AB'C + AB'C'$

$$= A \cdot [B + B'C + B'C']$$

$$= A \cdot [B + C + B'C']$$

$$= A \cdot [B + C' + C]$$

$$= A \cdot [B + 1]$$

$$= A \cdot 1$$

$$= A$$

$$\square T_a$$

$$\square T_b$$

$$\square T_c$$

$$T = T_a + T_b + T_c$$

$$T' = T_a + T_b \quad T > T'$$

Rules: -

1. Compliment: $(A')' = A$

2. AND:

$$A \cdot A = A$$

$$A \cdot 0 = 0$$

$$A \cdot 1 = A$$

$$\underline{A \cdot A' = 0}$$

3. OR:

$$A + A = A$$

$$A + 0 = A$$

$$\underline{A + 1 = 1}$$

$$A + A' = 1$$

4. Distributive:

$$A + BC = (A + B) \cdot (A + C)$$

$$A \cdot (B + C) = A \cdot B + A \cdot C$$

$$\begin{matrix} * & A + A'B = A + B \\ & \underline{A' + AB = A' + B} \end{matrix}$$

5. De Morgan's Law:

$$\begin{cases} (A+B)' = A' \cdot B' \\ (A \cdot B)' = A' + B' \end{cases}$$

+ NOT

x AND

OR

Boolean Algebra Examples (Part 2)

Ex: 3 $F = (A+B+C)(A+B'+C)(A+B+C')$

H.W.

$$Y = A \cdot B + A' \cdot C + B \cdot C$$

Rules: -

1. Compliment: $(A')' = A$

2. AND: $A \cdot A = A$

$$A \cdot 0 = 0$$

$$A \cdot 1 = A$$

$$A \cdot A' = 0$$

3. OR: $A+A = A$

$$A+0 = A$$

$$A+1 = 1$$

$$A+A' = 1$$

4. Distributive: $A+BC = (A+B) \cdot (A+C)$
 $A \cdot (B+C) = A \cdot B + A \cdot C$

$$* A+A'B = A+B$$

$$A'+AB = A'+B$$

5. De Morgan's Law:

$$(A+B)' = A' \cdot B'$$

$$(A \cdot B)' = A' + B'$$

Ex: 4 $G = (A+B)(A+B')(A'+B)(A'+B')$

Boolean Algebra Examples (Part 2)

Ex: 3 $F = (A+B+C)(A+B'+C)(A+B+C')$

Sol: - let $x = A+B$

$$F = (x+C) \cdot (A+B'+C) \cdot (x+C')$$

$$F = (x+C) \cdot (x+C') \cdot (A+B'+C)$$

$$F = (x+C \cdot C') \cdot (A+B'+C)$$

$$F = (x+0) \cdot (A+B'+C) \Rightarrow F = x \cdot (A+B'+C)$$

$$= (A+B) \cdot (A+B'+C)$$

$$= A+B \cdot (B'+C)$$

$$= A+B \cdot B' + B \cdot C$$

$$= A+0+B \cdot C \Rightarrow \underline{F = A+B \cdot C}$$

H.W.

$$\Rightarrow Y = A \cdot B + A' \cdot C + B \cdot C$$

$$A \cdot B = B \cdot A$$

$$A \cdot B \cdot C = A \cdot C \cdot B$$

Rules: -

1. Compliment: $(A')' = A$

2. AND: $A \cdot A = A$

$$A \cdot 0 = 0$$

$$A \cdot 1 = A$$

$$\leftarrow A \cdot A' = 0$$

3. OR: $A+A = A$

$$A+0 = A$$

$$A+1 = 1$$

$$A+A' = 1$$

4. Distributive: $A+BC = (A+B)(A+C)$
 $A(B+C) = AB + AC$

$$* A+A'B = A+B$$

$$A'+AB = A'+B$$

5. De Morgan's Law:

$$(A+B)' = A' \cdot B'$$

$$(A \cdot B)' = A' + B'$$

Ex: 4 $G = (A+B)(A+B')(A'+B)(A'+B')$

Ex: 4 $G = (A+B)(A+B')(A+B)(A+B')$

Sol: - $G = (A+B.B').(A'+B.B')$

$$G = (A+0).(A'+0)$$

$$G = A.A'$$

$$G = 0 \quad \underline{\text{Ans}}$$

Rules: -

1. Complement: $(A')' = A$

2. AND: $A.A = A$

$$A.0 = 0$$

$$A.1 = A$$

$$A.A' = 0$$

3. OR: $A+A = A$

$$A+0 = A$$

$$A+1 = 1$$

$$A+A' = 1$$

4. Distributive: $A+BC = (A+B).(A+C)$

$$A.(B+C) = A.B + A.C$$

$$* A+A'B = A+B$$

$$A'+AB = A'+B$$

5. De Morgan's Law:

$$(A+B)' = A'.B'$$

$$(A.B)' = A'+B'$$

Trick

Redundancy Theorem

Consensus theorem.

- i) Three variables.
- ii) Each variable is repeated twice.
- iii) One variable is complimented.
- iv) Take the complimented variable.

$$Y = \underline{AB} + \underline{A'C} + \underline{BC}$$

Redundant
term

$$Y = \underline{A \cdot B} + \underline{A' \cdot C}$$

$$Y = A \cdot B + A' \cdot C + BC \cdot 1$$

$$Y = A \cdot B + A' \cdot C + B \cdot C \cdot (A + A')$$

$$Y = \underline{A \cdot B} + \underline{A' \cdot C} + \underline{ABC} + \underline{A'BC}$$

$$Y = AB[1+C] + A'C[1+B]$$

$$Y = A \cdot B + A' \cdot C$$

- i) Three variables.
- ii) Each variable is repeated twice.
- iii) One variable is complimented.
- iv) Take the complimented variable.

$$i) F = \underline{AB} + \underline{B\bar{C}} + \underline{AC}$$

$$F = B\bar{C} + AC$$

u.w
ii)

$$F = A\bar{B} + BC + AC$$

$$iii) f = \underline{(A+B)} \cdot \underline{(\bar{A}+C)} \cdot \underline{(B+C)}$$

$$f = (A+B) \cdot (\bar{A}+C)$$

u.w
iv)

$$f = (A+B) \cdot (\bar{B}+C) \cdot (A+C)$$

$$v) F = \bar{A}\bar{B} + \underline{A\bar{C}} + \underline{\bar{B}C}$$

$$F = \bar{A}\bar{B} + A\bar{C}$$

$$Y = \underline{A \cdot B} + \underline{A' \cdot C}$$

$$Y = A \cdot B + A' \cdot C + BC \cdot 1$$

$$Y = A \cdot B + A' \cdot C + B \cdot C \cdot (A + A')$$

$$Y = \underline{A \cdot B} + \underline{A' \cdot C} + \underline{ABC} + \underline{A'BC}$$

$$Y = AB[1+C] + A'C[1+B]$$

$$Y = A \cdot B + A' \cdot C$$

(SOP form)

Sum of Product (Part 1)

total no. of comb. = $2^n \rightarrow$ no. of variables

$$= 2^3 = 8$$

M

A	B	C	F
			$\rightarrow$ SOP form
			$\rightarrow$ POS form

0 m_0	0	0	0
1 m_1	0	1	0
2 m_2	1	0	①
3 m_3	1	1	0
4 m_4	0	0	1
5 m_5	0	1	1
6 m_6	1	0	1
7 m_7	1	1	1

$$F(A, B, C) = m_2 + m_4 + m_5 + m_6 + m_7$$

$$= F(A, B, C) = \sum m(2, 4, 5, 6, 7)$$

written directly
 $\uparrow$ from T.T.

SOP form

$$F = \underbrace{\bar{A} \cdot \bar{B} \cdot \bar{C}}_{\text{AND op.}} + \underbrace{A \cdot \bar{B} \cdot \bar{C}}_{\text{OR}} + \underbrace{A \cdot \bar{B} \cdot C} + \underbrace{A \cdot B \cdot \bar{C}} + \underbrace{A \cdot B \cdot C}$$

Standard or
Canonical
SOP form

min term $\rightarrow \bar{A} \cdot B \cdot C$

$AB \cdot C$

AB

AC

$\bar{A}C$ etc.

$$(A+B+C)$$

$$A \cdot B \cdot C$$

AND sp.

$$F = \bar{A} \cdot B \cdot \bar{C} + \bar{A} \cdot \bar{B} \cdot \bar{C} + \bar{A} \cdot \bar{B} \cdot C + \bar{A} \cdot B \cdot \bar{C} + \bar{A} \cdot B \cdot C$$

m2 m4 m5 m6 m7

$$F = \bar{A} B \bar{C} + \bar{A} \bar{B} [\bar{C}/C] + \bar{A} B [\bar{C}/C]$$

$$= \bar{A} B \bar{C} + \bar{A} \bar{B} + \bar{A} B$$

$$\bar{C} + C = 1$$

$$= \bar{A} B \bar{C} + A [\bar{B} + B] \Rightarrow F = \bar{A} B \bar{C} + A$$

$$A + (\bar{A})B$$

$$A + (\bar{A})X$$

$$F = A + X$$

$$F = A + B \bar{C} \quad \text{SOP}$$

minimal SOP form

Canonical / standard SOP

using minimal SOP

$$F = A + B \bar{C}$$

Sum of Product (Part 2)

Canonical/Standard SOP form: Each minterm is having all the variables in normal or complimented form.

$$Ex: - F = \bar{A}B + AB + \bar{A}\bar{B}$$

Minimal SOP form: Each minterm does not have all the variables in normal or complimented form.

$$Ex: - G = A + \bar{B}C$$

Que: For the given truth table, minimize the SOP expression

A	B	Y
0	0	0
0	1	1
1	0	0
1	1	1

Que: Simplify the expression for-
 $Y(A,B) = \sum m(0,2,3)$

Que: For the given truth table, minimize the SOP expression

	A	B	Y
m_0	0	0	0
m_1	0	1	1
m_2	1	0	0
m_3	1	1	1

$$1 \rightarrow A$$

$$0 \rightarrow \bar{A}$$

directly written from T.T.

Sol:-

$$Y = \bar{A} \cdot B + A \cdot B \quad \text{canonical / st. SOP form}$$

$$Y = (\bar{A} + A) \cdot B$$

$$Y = 1 \cdot B \Rightarrow \boxed{Y = B} \quad \text{minimal SOP form}$$

$$\begin{aligned} &= \bar{A} + \textcircled{A} \cdot B \\ &= \bar{A} + B \end{aligned}$$

Que: Simplify the expression for-
 $Y(A,B) = \sum m(0,2,3)$

$$Y = m_0 + m_2 + m_3$$

$$Y = \bar{A} \bar{B} + A \cdot \bar{B} + A \cdot B$$

Can. SOP

$$Y = (\bar{A} + A) \cdot \bar{B} + A \cdot B$$

$$= 1 \cdot \bar{B} + A \cdot B$$

$$= \bar{B} + \textcircled{A \cdot B}$$

$$Y = \bar{B} + A \quad \text{min SOP}$$

$$2^3 = 8$$

Product of Sum (POS Form)

A	B	C	Y
0	0	0	0
0	0	1	0
0	1	0	1
0	1	1	0
1	0	0	1
1	0	1	1
1	1	0	1
1	1	1	1

→ POS form is used when the o/p is "0"

$$\begin{array}{c|c} 0 \rightarrow \bar{A} & 0 \rightarrow \bar{A} \\ \hline 1 \rightarrow \bar{A} & 1 \rightarrow A \\ \hline \text{POS} & \text{SOP} \end{array}$$

$$(A+B+C) =$$

$$(A \cdot B \cdot C) =$$

$$Y = (A+B+C) \cdot (A+B+\bar{C}) \cdot (A+\bar{B}+\bar{C})$$

maximum

$$\bar{Y} = \bar{A} \cdot \bar{B} \cdot \bar{C} + \bar{A} \cdot \bar{B} \cdot C + \bar{A} B C$$

0	1	1	0
1	0	0	1
1	0	1	1
1	1	0	1
1	1	1	1

$$(A')' = A$$

$$Y = (A+B+C) \cdot (A+B+\bar{C}) \cdot (A+\bar{B}+\bar{C})$$

maximum

$$\bar{Y} = \bar{A} \cdot \bar{B} \cdot \bar{C} + \bar{A} \cdot \bar{B} \cdot C + \bar{A} B C$$

complement of both sides

$$(\bar{Y}) = [\bar{A} \bar{B} \bar{C} + \bar{A} \bar{B} C + \bar{A} B C]$$

$$Y = (\bar{A} \bar{B} \bar{C})' \cdot (\bar{A} \bar{B} C)' \cdot (\bar{A} B C)'$$

$$Y = (A+B+C) \cdot (A+B+\bar{C}) \cdot (A+\bar{B}+\bar{C})$$

canonical
pos form

$$\overline{(A+B)} = A' \cdot B'$$

$$Y = \frac{(A+B+C) \cdot (A+B+\bar{C})}{X} \cdot \frac{(A+\bar{B}+\bar{C})}{X}$$

$$Y = (X+C\bar{C}) \cdot (A+\bar{B}+\bar{C})$$

$$Y = (A+B)(A+\bar{B}+\bar{C})$$

$$Y = A + B \cdot (\bar{B} + \bar{C}) \Rightarrow Y = A + B \cdot \bar{B} + B \bar{C} \quad \nearrow$$

$$Y = A + B \bar{C} \Rightarrow Y = \underline{(A+B)} \cdot \underline{(A+\bar{C})}$$

$$(A+B) \cdot (A+C) = A+BC$$

$$(X+C) \cdot (X+\bar{C}) = X+C\bar{C}$$

minimal POS form

Product of Sum (POS Form)

→ canonical form
→ minimal form

Que: For the given truth table minimize the POS expression.

	A	B	Y
M_0	0	0	1
M_1	0	1	0
M_2	1	0	1
M_3	1	1	0

max term

$$Y = (A + \bar{B}) \cdot (\bar{A} + \bar{B})$$

$$Y = \bar{B} + A \cdot \bar{A}$$

$$Y = \bar{B} + 0$$

$$Y = \bar{B} \text{ — minimal form}$$

$$0 \rightarrow A \quad Y = \overline{AB} + \overline{\bar{A}B}$$

AND AND

$$1 \rightarrow \bar{A}$$

standard POS form

$$(A+B) \cdot (A+C) = A+BC$$

Σ

SOP Form: — AND

POS Form: — OR

$$Y = \prod(M_1, M_3)$$

$$Y = \prod M(1, 3)$$

$$Y = \sum m(0, 2)$$

SOP & POS Form Example

A	B	C	Y
0	0	0	<u>1</u>
0	0	1	<u>0</u>
0	1	0	<u>1</u>
0	1	1	<u>1</u>
1	0	0	<u>0</u>
1	0	1	<u>0</u>
1	1	0	<u>1</u>
1	1	1	<u>1</u>

SOP form: - $\begin{cases} 0 \rightarrow \bar{A} \\ 1 \rightarrow A \end{cases}$
 $\bar{Y} \Rightarrow 0$
 $Y \Rightarrow 1$

or $Y(A, B, C) = \sum m(0, 2, 3, 6, 7)$

1, 4, 5

$$Y(A, B, C) = \prod M(1, 4, 5)$$

$$Y = \bar{A} \cdot \bar{B} \cdot \bar{C} + \bar{A} \cdot B \cdot \bar{C} + \bar{A} \cdot B \cdot C + A \cdot B \cdot \bar{C} + A \cdot B \cdot C$$

Canonical SOP form

$$A + (\overline{A})B = \underline{A+B}$$

$$\hookrightarrow (\underbrace{A + \overline{A}}_1) \cdot \underline{(A+B)}$$

$$Y = \overline{A}\overline{B}\overline{C} + \overline{A}B(\overline{C} + \underbrace{C}_1) + AB(\overline{C} + C)$$

$$= \overline{A}\overline{B}\overline{C} + \overline{A} \cdot B \cdot 1 + AB \cdot 1$$

$$= \overline{A}\overline{B}\overline{C} + B(\overline{A} + A)$$

$$= \underline{\overline{A}\overline{B}\overline{C}} + B \quad \text{let } X = \overline{A}\overline{C}$$

$$= X \cdot (\underline{B}) + B$$

$$= X + B \Rightarrow \boxed{Y = \overline{A}\overline{C} + B}$$

Minimal SOP form

POS form:-
 $0 \rightarrow A$
 $1 \rightarrow \bar{A}$

$$Y = (A+B+\bar{C}) \cdot (\bar{A}+B+C) \cdot (\bar{A}+B+\bar{C})$$

Canonical POS form

$$Y = (A+B+\bar{C}) \cdot \left(\frac{X}{\bar{A}+B+C \cdot \bar{C}} \right) \quad X+0 = X$$

$$Y = (A+B+\bar{C}) \cdot (\bar{A}+B)$$

$$Y = B + (A+\bar{C}) \cdot \bar{A}$$

$$= B + \underset{\substack{\swarrow \\ 0}}{A/\bar{A}} + \bar{A} \cdot \bar{C} \Rightarrow Y = B + \bar{A} \cdot \bar{C} \Rightarrow Y = \frac{(\bar{A}+B) \cdot (B+\bar{C})}{\text{minimal POS form}}$$

Minimal to Canonical Form Conversion (Part 1)

$$Y = \underline{A + B'C}$$

$$Y = A \cdot \underset{C+0}{\underset{B+B'}{11}} + B'C \cdot \underset{A+A'}{1}$$

$$= A \cdot (B+B') \cdot (C+0) + B'C \cdot (A+A')$$

$$= (A \cdot B + \underline{A \cdot B'}) \cdot (C+0) + \underline{B'C \cdot A + B'C \cdot A'}$$

$$Y = A \cdot B \cdot C + A \cdot B \cdot 0 + \underline{A \cdot B' \cdot C + A \cdot B' \cdot 0} + \underline{\cancel{B'C} \cdot A} + B'C \cdot A'$$

$$Y = ABC + ABC' + AB'C + AB'C' + A'B'C$$

Ans

Step:- 1 3 variables

A, B & C

Step:- 2

M ₁	A ✓	M ₂	A ✗
	B ✗		B ✓
	C ✗		C ✓

Step:- 3

SOP $\overline{A} \overline{B}$

$$= \textcircled{M_1} + \textcircled{M_2} + \textcircled{M_3}$$

↓
minimal

$$g = ABC + AB'C$$

$$= AC (B + B')$$

$$= AC \cdot 1$$

$$g = \underline{AC}$$

Minimal to Canonical Form Conversion (Part 2)

all variables
↑ are present

$F = (A+B+C')(A'+C)$ minimal POS

$F = (A+B+C')(A'+C+0)$

$= (A+B+C')(\underbrace{A'+C}_x + B \cdot B')$

$= (A+B+C')(\underbrace{x + B \cdot B'}_y)$

$= (A+B+C') \cdot (x+B) \cdot (x+B')$

$F = (A+B+C') \cdot (A'+C+B) \cdot (A'+C+B')$

Canonical POS

POS

i) A, B, C

Term 1

A ✓
B ✓
C ✓

Term 2

A ✓
B x
C ✓

iii)

$= (x + y \cdot y')$

$= \underline{(x+y) \cdot (x+y')}$

$G = (A+B) \cdot (A'+C)$
 $= (A+B+\underbrace{0}_{C \cdot C'}) \cdot (A'+C+\underbrace{0}_{B \cdot B'})$

Examples & Tricks (SOP & POS Forms)

Ex: In a minimal SOP form, number of minterms in the logical expression $A + B'C$ are:

a) 4

b) 6

☒ c) 5

d) 7

$$F = \underbrace{A}_{T1} + \underbrace{B'C}_{T2}$$

$$\begin{array}{ll} \hookrightarrow A \checkmark & \hookrightarrow A \times \\ \hookrightarrow B \times & \hookrightarrow B \checkmark \\ \hookrightarrow \bar{C} \times & \hookrightarrow C \checkmark \end{array}$$

3 variables

A, B & C

$$A + A = A$$

$$F = A \cdot 1 \cdot 1 + B'C \cdot 1$$

$$f = A \cdot (B + \bar{B}) \cdot (C + \bar{C}) + B'C \cdot (A + \bar{A})$$

$$F = (AB + A\bar{B}) \cdot (C + \bar{C}) + \overbrace{AB'C + A'B'C}$$

$$f = ABC + ABC' + \underline{AB'C} + \boxed{A'B'C}$$

$$F = \underset{1}{ABC} + \underset{2}{ABC'} + \underset{3}{AB'C} + \underset{4}{AB'C'} + \underset{5}{A'B'C}$$

Ex: With 2 variables maximum possible minterms and maxterms are:

a) 2

b) 4

c) 8

d) 16

$$Y=1$$

$$Y=0$$

Ex: With 2 variables maximum possible minterms and maxterms are:

a) 2

~~b) 4~~

c) 8

d) 16

n variables

$$2^n = \text{mint / maxterm}$$

(A, B)

$$2^2 = 4$$

		Y	
		0	1
		0	1
0	0	0	0
0	1	0	1
1	0	1	0
1	1	1	1

Ex: For $n=4$, what is the total number of logical expression:

Ex: For $n=4$ what is the total number of logical expression:

imp $\rightarrow$ no. of variables

for $n=2$ (A, B)

1	A	$A\bar{B}$	$A\bar{B} = \bar{B}A$
0	$\bar{A}$	$\bar{A}B$	$A+B = B+A$
$\bar{A}B + A\bar{B}$	B	$A+\bar{B}$	$\bar{A}\bar{B}$
$AB + \bar{A}\bar{B}$	$\bar{B}$	$\bar{A}+B$	$\bar{A}+\bar{B}$

16 logical exp.

$$16 = 2^4 \rightarrow \text{variable}$$
$$16 = 2^{2^2}$$

$$2^{2^n}$$

$$2^4$$

$$= 2^{16} = 65536$$

ans

Positive and Negative Logic

"Positive Logic"

↑ voltage corres. to
logic "1"

→ "5V" ⇒ logic "1"

"0V" ⇒ logic "0"

Ex:- logic 0 → -5V
logic 1 → 0V
positive logic

"Neg. Logic"

↑ voltage correspond to
logic "0"

→ 5V ⇒ logic "0"

0V ⇒ logic "1"

0	>	-5
'1'	'0'	P.L.
'0'	'1'	N.L.

"Dual form"

ex:- logic 0 → -1.7V
logic 1 → -0.8V

Dual Form

- Positive and Negative logic AND gate.
- Positive and Negative logic OR gate.
- What is Dual form?
- Rules to write Dual form.

↑ → "1"
↓ → "0"

⊕^{ve} logic AND Gate

A	B	Y
0	0	0
0	1	0
1	0	0
1	1	1

A	B	Y
L	L	L
L	H	L
H	L	L
H	H	H

⊖^{ve} logic AND Gate

A	B	Y
1	1	1
1	0	1
0	1	1
0	0	0

A	B	Y
L	L	L
L	H	H
H	L	H
H	H	H

↓
 $\oplus^{\vee}$ logic OR Gate

A	B	Y
0	0	0
0	1	1
1	0	1
1	1	1

$\ominus^{\vee}$ logic OR Gate

A	B	Y
1	1	1
1	0	0
0	1	0
0	0	0

$$\begin{array}{l} \underbrace{A \cdot B}_{\oplus^{\vee}} \xrightarrow{\text{Dual}} A + B \\ A + B \xrightarrow{\oplus^{\vee}} A \cdot B \end{array}$$

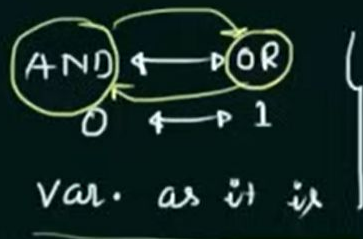
$\oplus^{\vee}$ logic AND Gate = $\ominus^{\vee}$ logic OR Gate

$\ominus^{\vee}$ logic — = $\oplus^{\vee}$ —

Self Dual

For any logical expression, two times dual gives the same expression.

» In self dual expression one time dual gives the same expression.



$$F = A \cdot B \cdot \bar{C} + \bar{A} \cdot B \cdot C + A \cdot B \cdot C$$

$$F' = (A+B+\bar{C}) \cdot (\bar{A}+B+C) \cdot (A+B+C)$$

$$F'' = A \cdot B \cdot \bar{C} + \bar{A} \cdot B \cdot C + A \cdot B \cdot C$$

$$G = A \cdot B + B \cdot C + A \cdot C$$

$$G' = \frac{(A+B) \cdot (B+C) \cdot (A+C)}{X+X=X}$$

$$= (B+A \cdot C) \cdot (A+C)$$

$$= A \cdot B + A \cdot A \cdot C + B \cdot C + A \cdot C \cdot C$$

$$= A \cdot B + \underline{A \cdot C} + B \cdot C + \underline{A \cdot C}$$

$$G' = A \cdot B + B \cdot C + A \cdot C$$

** n variables $\Rightarrow 2^{2^n - 1}$

$$n = 2$$

$$\begin{aligned} \text{no. of S.O. exp} &= 2^{2^2 - 1} \\ &= 2^{2^1} \\ &= 4 \end{aligned}$$

A & B

A $\rightarrow$ A

B $\rightarrow$ B

$\bar{A} \rightarrow \bar{A}$

$\bar{B} \rightarrow \bar{B}$

H.W.

$$1) Y = \bar{A} \cdot \bar{B} + \bar{B} \cdot \bar{C} + \bar{A} \cdot \bar{C}$$

$$2) n = 4$$

Complement Meaning and Examples

$$U = \{a, e, i, o, u\}$$

$$\underline{A = \{a, e, i\}}$$

$$\bar{A} \text{ or } A'$$

$$\bar{A} = U - A \Rightarrow U = A + \bar{A}$$

$$\bar{A} = \{o, u\} = \{a, e, i\} + \{o, u\} \quad \underline{0 \leftrightarrow 1}$$

$$U = \{a, e, i, o, u\}$$

$$A \begin{matrix} \rightarrow 0 \\ \rightarrow 1 \end{matrix}$$

$$U = \{0, 1\}$$

$$A = \{\underline{0}\} \quad | \quad B = \{\underline{1}\}$$

$$\bar{A} = \{\underline{1}\} \quad | \quad \bar{B} = \{\underline{0}\}$$

$$F = \bar{A}B + \bar{B}A$$

$$\bar{F}?$$

$$\left\{ \begin{array}{l} \text{AND} \leftrightarrow \text{OR} \\ \cdot \leftrightarrow + \\ 0 \leftrightarrow 1 \end{array} \right. \quad \text{comp of each var.}$$

H.W.
Ex. -

$$G = ABC + \bar{A}BC + A\bar{B}C$$

$\bar{G}?$

$$\bar{F} = \overline{(\bar{A}B + \bar{B}A)}$$

$$\bar{A}B = L \quad \& \quad \bar{B}A = M$$

$$\bar{F} = \overline{(L + M)}$$

$$= \bar{L} \cdot \bar{M}$$

$$= \overline{(\bar{A} \cdot B)} \cdot \overline{(\bar{B} \cdot A)}$$

$$\bar{F} = (A + \bar{B}) \cdot (B + \bar{A}) \quad \text{Ans}$$

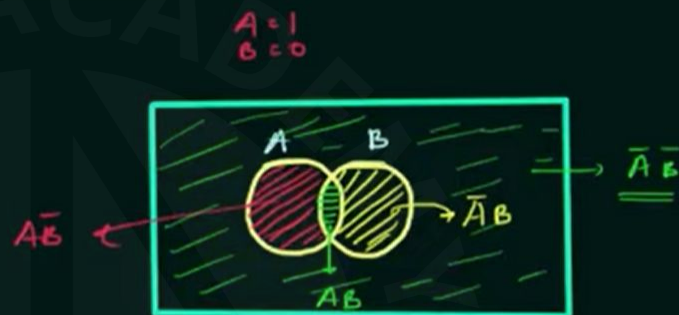
$$= A \cdot B + \underbrace{A/\bar{A}}_0 + \underbrace{\bar{B}/B}_0 + \bar{B} \cdot \bar{A}$$

$$\bar{F} = \underline{A \cdot B + \bar{A} \bar{B}} \quad \text{Ans}$$

$A \odot B$

Venn Diagram

Two Variables (A, B)



Minimize the SOP expression for shaded region.

$$Y = \bar{A}\bar{B} + A\bar{B} + AB$$

$$Y = (\bar{A} + A) \cdot \bar{B} + AB$$

$$Y = \bar{B} + AB \Rightarrow Y = (\bar{B} + A) \cdot (\bar{B} + B)$$

$$Y = A + \bar{B} \quad \text{Ans}$$



(a)

$$G = \overline{A} \cdot \overline{B} + A \cdot \overline{B} + A \cdot \underline{B} + \overline{A} \cdot \underline{B}$$

$$G = (\overline{A} + A) \overline{B} + (A + \overline{A}) \cdot B$$

$$G = \overline{B} + B$$

$$G = 1$$



(b)

a) $(AB + BC + CA) \overline{E}$

b) $(A\overline{B}\overline{C} + \overline{A}B\overline{C} + A\overline{B}C) \overline{E}$

c) $(A\overline{B}C + \overline{A}\overline{B}\overline{C}) \overline{E}$

d) $\overline{C}\overline{B}\overline{A} + A\overline{B}\overline{C} + \overline{E}$



(c)

Switching Circuit (Part 1)

1 sw $\rightarrow$ 2 comb.

2 sw $\rightarrow$ 4 comb

$M \rightarrow$ sw

$n \rightarrow$ comb

$$2^M = n \quad \text{or} \quad M = \log_2 n$$

$$n = 8$$

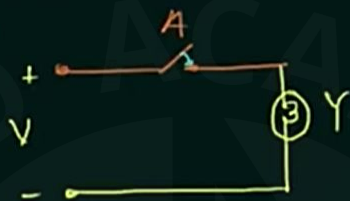
$$M = \log_2 8$$

$$= \log_2 2^3$$

$$= 3 \times \frac{1}{1} \times \frac{1}{2}$$

$$M = 3$$

Series;



A	Y
(OFF) 0	0
(ON) 1	1

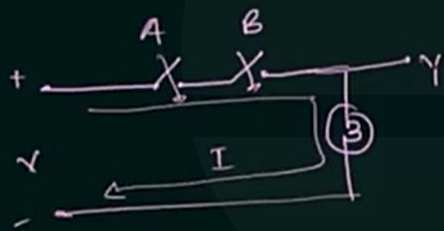
parallel; NOT



A	Y
0	1
1	0

AND :

A	B	Y (bulb)
0	0	0
0	1	0
1	0	0
1	1	1



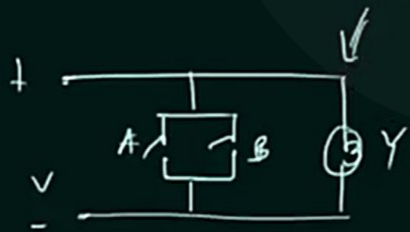
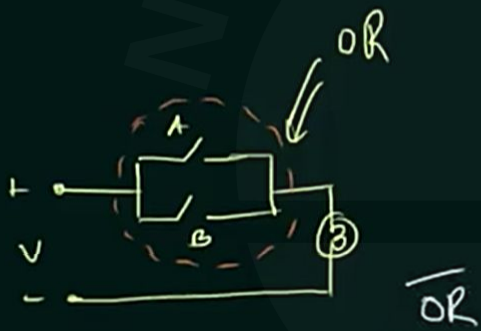
OR :

A	B	Y
0	0	0
0	1	1
1	0	1
1	1	1



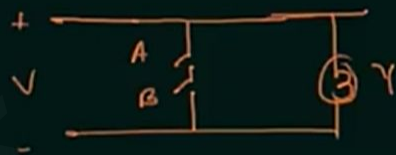
NOR: $Y = \overline{A+B}$

A	B	Y
0	0	1
0	1	0
1	0	0
1	1	0



NAND: $Y = \overline{A \cdot B}$

A	B	Y = $\overline{A \cdot B}$
0	0	1
0	1	1
1	0	1
1	1	0

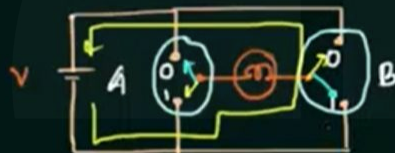
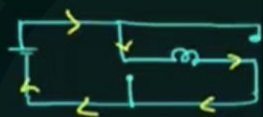


Switching Circuit (Part 2)

1. EXOR "odd 1's detector"

A	B	$Y = A'B + BA' = A \oplus B$
0	0	0
0	1	1
1	0	1
1	1	0

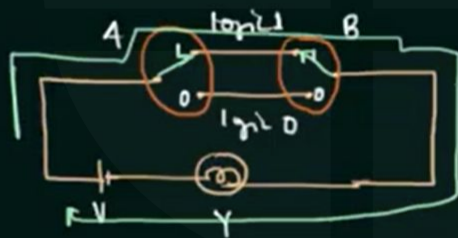
A	B	C	D	Y
1	0	1	1	1



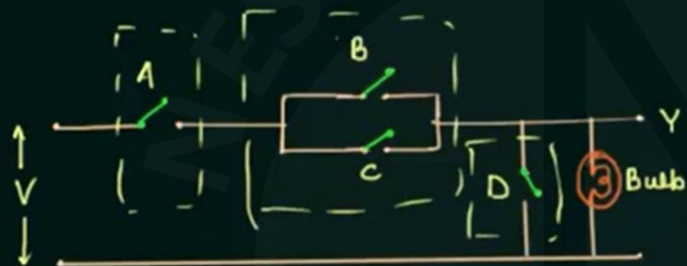
Ex - NOR:

$$Y = A'B + AB = A \odot B$$

A	B	Y
0	0	1
0	1	0
1	0	0
1	1	1



Ex: Determine Y.



Sol: $Y = \underline{A \cdot (B + C) \cdot \bar{D}}$

$$\begin{aligned} Y &= \underline{A \cdot B \cdot \bar{D}} + \underline{A \cdot C \cdot \bar{D}} \\ &= A \cdot (B + C) \cdot \bar{D} \end{aligned}$$

Statement Problem (Part 1)

1. Consider a logic circuit with three inputs A, B and C . Output Y is 1 (high) for the following conditions:

- i) B and C are true.
- ii) A and C are false.
- iii) A, B and C are true.
- iv) A, B and C are false.

Minimize the function Y .

Statement Problem (Part 1)

1. Consider a logic circuit with three inputs A, B and C. Output Y is 1 (high) for the following conditions:

- =i) B and C are true. $B \cdot C$
- =ii) A and C are false. $A' \cdot C'$
- =iii) A, B and C are true. $A \cdot B \cdot C$
- =iv) A, B and C are false. $A' \cdot B' \cdot C'$

Minimize the function Y.

Solution:
$$Y = \underline{B \cdot C} + \underline{A' \cdot C'} + A \cdot \underline{B \cdot C} + \underline{A' \cdot B' \cdot C'}$$

$$Y = BC(1 + A) + A' \cdot C'(1 + B')$$

$$Y = BC \cdot 1 + A' \cdot C' \cdot 1$$

$$\therefore Y = BC + A' \cdot C'$$

2. Y for the con.

=i) A & B are T

=ii) $A \rightarrow T$
 $B \rightarrow F$
 $C \rightarrow T$

=iii) $A \rightarrow T$
 $B \rightarrow F$
 $C \rightarrow F$

Statement Problem

3. A logic circuit have three inputs A,B & C. The output F is high when the majority of inputs are logic 1.
- a) Minimize the function.
 - b) Implement the circuit.

Solution

A	B	C	F
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	1
1	1	1	1

$$F = A'B'C + AB'C + \underline{ABC'} + \underline{ABC}$$

$$F = A'B'C + AB'C + AB(\cancel{C'} + C)$$

$$F = A'B'C + \underline{AB'C} + \underline{AB}$$

$$F = A'B'C + A(B'C + B)$$

$$F = A'B'C + A(C + B)$$

$$F = \underline{A'B'C} + \underline{AB} + AC$$

$$F = (A'C + A)B + AC$$

$$F = (C + A)B + AC$$

$$F = \underline{AB + AC + BC} \quad \text{Ans}$$

$$X + X' = 1$$

$$AB \cdot 1 = AB$$

$$\underline{X'Y + X = (X + Y)}$$

$$F = A'BC + AB'C + \underline{ABC'} + \underline{ABC}$$

$$F = A'BC + AB'C + AB(\cancel{C'} \downarrow \underset{1}{C})$$

$$F = A'BC + \underline{AB'C} + \underline{AB}$$

$$F = A'BC + A(B'C + B)$$

$$F = A'BC + A(C + B)$$

$$F = A'\underline{BC} + A\underline{B} + AC$$

$$F = (A'C + A)B + AC$$

$$F = (C + A)B + AC$$

$$\textcircled{f} = \underline{\textcircled{AB} + \textcircled{AC} + \textcircled{BC}} \quad A_{12}$$

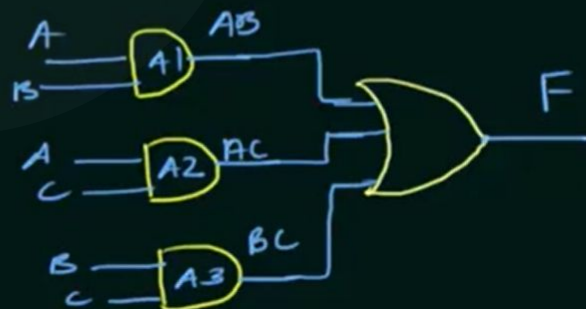
$$X + X' = 1$$

$$AB \cdot 1 = AB$$

$$\underline{X'Y + X = (X + Y)}$$

3 AND

1 OR (3 I/P)





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